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Ofloxacin resistance in multibacillary new leprosy cases from Purulia, West Bengal: a threat to effective secondary line treatment for rifampicin-resistant leprosy cases

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A B S T R A C T

Objectives: Purulia is one of the high-endemic districts for leprosy in West Bengal (the eastern part of India). The annual new case detection rate (ANCDR) of leprosy in West Bengal is 6.04/100000 (DGHS 2019–20). Our earlier report provided evidence of secondary drug resistance in relapse cases of leprosy. The aim of the current study was to observe primary drug resistance patterns for dapsone, rifampicin, and ofloxacin amongst new leprosy patients from Purulia, West Bengal in order to better understand the emergence of primary resistance to these drugs.
Methods: In the present study, slit-skin smear samples were collected from 145 newly diagnosed leprosy cases from The Leprosy Mission (TLM) Purulia hospital between 2017 and 2018. DNA was extracted from these samples and the *Mycobacterium leprae* genome was analyzed for genes associated with drug resistance by polymerase chain reaction (PCR), followed by Sanger sequencing. Wild-type strain (Thai-53) and mouse footpad-derived drug-resistant strain (Z-4) were used as reference strains.
Results: Of 145 cases, 25 cases showed mutations in genes associated with resistance to rifampicin, dapsone, and ofloxacin (as described by the World Health Organization, *rpoB, folP*, and *gyrA*, respectively) through Sanger sequencing. Of these 25 cases, 16 cases showed mutations in ofloxacin, two cases showed mutations in combinations of ofloxacin and rifampicin, four cases showed a mutation only in rifampicin, one case showed mutations in combinations of rifampicin and dapsone, and two cases showed mutations only in dapsone.
Conclusion: Results from this study indicated the emergence of resistance to antileprosy drugs in new cases of leprosy. As ofloxacin is the alternate drug for the treatment of rifampicin-resistant cases, the emergence of new cases with resistance to ofloxacin indicates that ofloxacin-resistant *M. leprae* strains are actively circulating in this endemic region (i.e., Purulia, West Bengal), posing challenges for the effective treatment of rifampicin-resistant cases.

1. Introduction

For the last three decades, the treatment and control of leprosy has been based on World Health Organization (WHO)-recommended multidrug therapy (MDT). After the introduction of WHO-recommended MDT, the prevalence of leprosy was drastically reduced around the world. However, many endemic pockets still exist. A total of 159 countries and territories report over 200,000 new leprosy cases annually [1,2], and India has reported approximately 56.6% of these cases (contributing 114,451 new cases) [3]. First-line drugs used in MDT for leprosy include dapsone, rifampicin, and clofazimine. The emergence of cases of rifampicin

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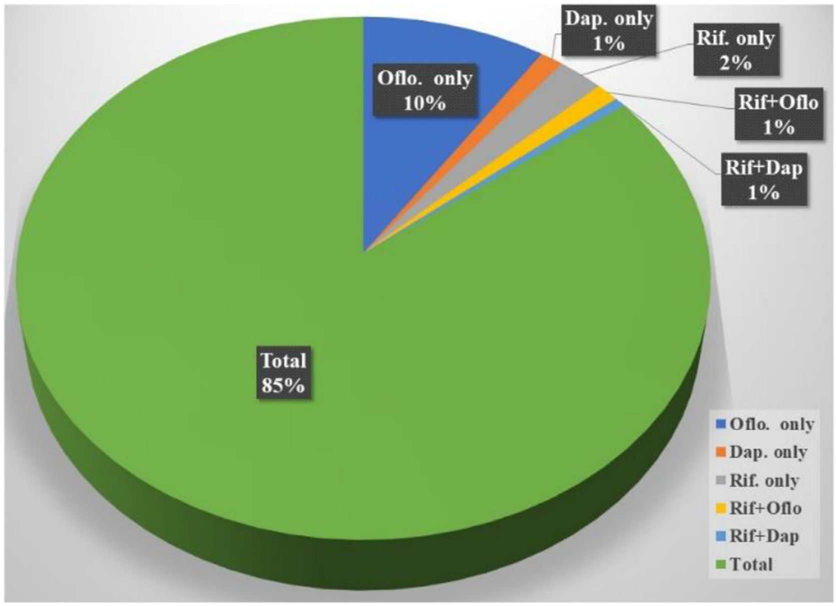

Fig. 1. Percentage of drug resistance among new leprosy cases.

resistance around the world may enable further transmission of rifampicin-resistant *Mycobacterium leprae* strains in endemic communities. To stop further transmission of rifampicin-resistant *M. leprae* strains, we need effective alternate drugs for the management of resistant cases. Recently, along with a few reports from different parts of the world, WHO recommended a second-line drug regimen for patients resistant to rifampicin and dapsone. This alternate regimen includes ofloxacin, minocycline, and clarithromycin [4].

The prevalence rate of leprosy is declining in India, but not rapidly enough. From 2019 to 2020, 114,451 new leprosy cases were detected in the country, accounting for 80% of cases reported in Southeast Asian countries and 56.6% of cases reported globally. The states in India that contributed more than 76% of the new leprosy cases were: Bihar, Maharashtra, Uttar Pradesh, Odisha, Chhattisgarh, Madhya Pradesh, West Bengal, and Jharkhand [5]. As of March 2018, 53 districts in 11 states/Union TerritoriesTs in India reported a prevalence rate (PR) >2/10,000 population, including West Bengal. The annual case detection rate (ANCDR) of West Bengal was 9.52/100,000 from 2017 to 2018 and came down to 6.04/100,000 from 2019 to 2020 [5]. Purulia is one of the endemic districts in West Bengal with an ANCDR of 3.0/100000 [5]. The present study was carried out in The Leprosy Mission (TLM) hospital in Purulia.

Owing to its high bactericidal property (99.9% elimination of *M. leprae*), rifampicin is the drug of choice for the treatment of leprosy [6]. According to the Global Sentinel Surveillance for Drug Resistance in Leprosy program coordinated by WHO, strains of *M. leprae* resistant to antileprosy drugs have been observed in several leprosy-endemic regions [7]. Recently, several reports of rifampicin-resistant cases from different parts of the world raised concerns, and effective treatments with other antibiotics were utilized to reduce transmission of drug-resistant strains in the community [8–11]. As recommended by WHO, ofloxacin is one of the main second-line drugs for the treatment of patients with rifampicin-resistant leprosy [4]. To control leprosy, it is essential, especially in high endemic pockets, to monitor drug sensitivity patterns of both rifampicin and ofloxacin. In our previous studies [10,11], we reported the existence of rifampicin-resistant cases in Purulia. As per WHO protocol, clinicians started prescribing second-line drugs for rifampicin-resistant cases to stop the transmission of resistant strains; for better management of resistant cases, we must monitor whether new patients are getting infections from resistant or nonresistant strains. The present study focused on the prevalence of drug (rifampicin, ofloxacin, and dapsone)-resistant strains of *M. leprae* in new patients with leprosy in the Purulia district of West Bengal.

2. Materials and methods

This study was approved by the Institutional Ethical Committee based on the guidelines of the Indian Council of Medical Research, New Delhi. Written informed consent was taken from all the patients recruited for this study. Skin slit scrapings were collected from new cases of leprosy from patients recruited at The Leprosy Mission Community Hospital, Purulia from 2017 to 2018. Skin slit skin scrapings were collected from 145 multibacillary (MB) leprosy cases in 0.5 mL Eppendorf tubes containing 70% ethanol. *Mycobacterium leprae* DNA was extracted using reagents of DNeasy Blood & Tissue Kits (Qiagen, Catalogue No: 69506) according to the manufacturer's instructions.

In alignment with WHO's Sentinel Surveillance for Drug Resistance in Leprosy program, we followed guidelines for the detection of mutations in the drug-resistance–determining regions (DRDRs) of the *M. leprae* genome, as well as guidelines for conventional polymerase chain reaction (PCR)-based gene amplification. [4]. The PCR mixture contained 12.5 μL of Hotstar Taq polymerase master mix (2X, Qiagen), forward and reverse primers at a final concentration of 0.5 mm, and 5 μL of template DNA. The final volume of reaction mix was made up to 25 μL with nuclease-free water. After PCR, the amplified fragment was detected using gel electrophoresis with ethidium bromide staining. Amplicons were excised from the gel and purified using Qiagen Gel Extraction Kit. (Qiagen, Germantown, MD). Eluted products were outsourced for sequencing (Eurofin Pvt Ltd., Bangalore). Sequenced data were analyzed using MEGA_X_10.1.8.

Amongst 145 samples, 25 (17.2%) were confirmed for the presence of drug-resistant strains (either resistance to rifampicin, ofloxacin, dapsone, or combinations of these), and the rest of the samples (120, 82.8%) were sensitive to all three drugs. Ofloxacin, dapsone, and rifampicin monodrug resistance was found in 16

Table 1
Clinical, geographical, and molecular details of all resistant cases of leprosy

S. No.	Type of leprosy	BI	Block	Mutation (position) in		
				gyrA (ofloxacin resistance)	*folp1* (dapsone resistance)	*rpoB* (rifampicin resistance)
1	LL	5	B.bazar	**G89F**	S	S
2	LL	4.33	Kotshila	**A91V**	S	S
3	BL	4	Adra	S	**A53T**	S
4	BL	3+	Puncha	**A91P**	S	S
5	BL	3.66	B.Bazar	**A89G**	S	**N458L**
6	BL	4	Boram	**A89G**	S	S
7	BL	4	Jhalda – i,	**T89G**	S	**Leu436Gln**
8	BL	3	Nimdih	**A89G**	S	S
9	BL	4	Santaldih	**L91A**	S	S
10	LL	4	B.bazar	**R91A**	S	S
11	LL	4	Prl-1	**A89G, L91A**	S	S
12	BL	2.66	Kenda-manbr-1	**A89G**	S	S
13	BL	4	Santaldih	**A89G, P91A**	S	S
14	LL	4.33	Prl-1	**A89G, F91A**	S	S
15	BL	3.66	M.bazar	S	S	**L436Q**
16	LL	5	Bagmundi	S	S	**A438Q, G441D, P451H, V456H, V458L**
17	BL	2	Giridih	S	**A53T**	**Leu436Gln**
18	BL		Bandwan	S	S	**L436Q**
19	BL		Ranchi	**A91V**	S	S
20	BL	5	Puncha	**A91V**	S	S
21	LL	5	Bugmundih	S	**A53T**	S
22	LL	5	Hura	**A89G**	S	S
23	BL	4.3	Chandankiary	S	S	**Leu458Pro**
24	BL	4	Muri	**A89G**	S	S
25	BL	1	Puncha	**A89G**	S	S

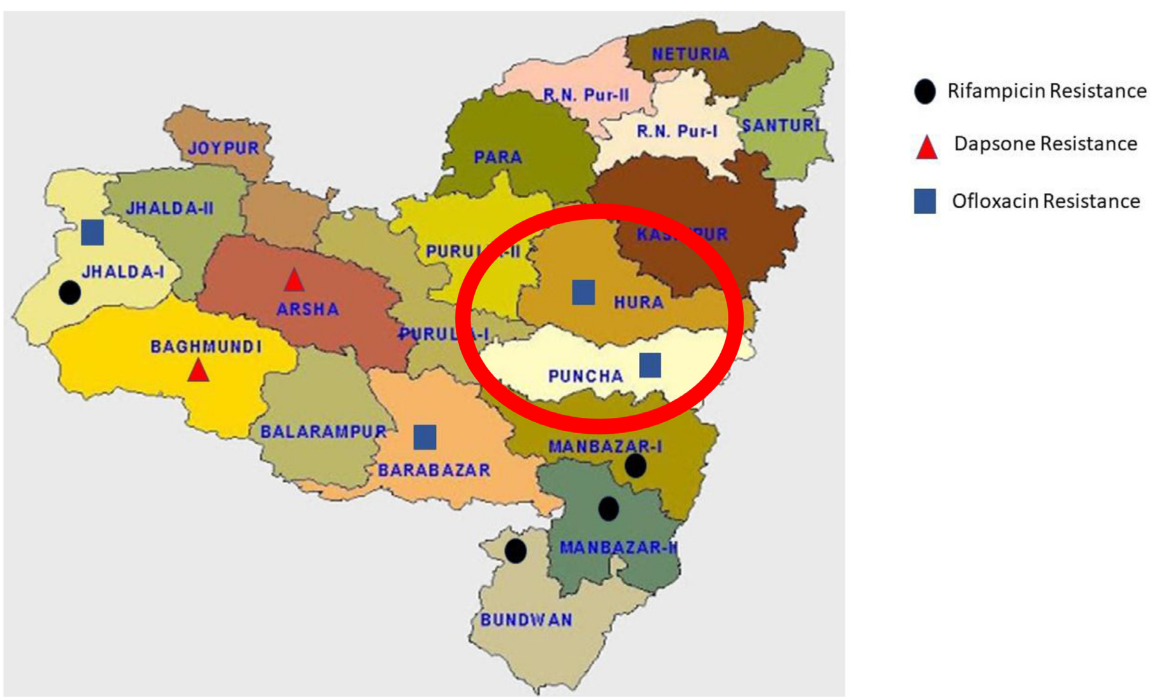

Fig. 2. Map of Purulia. Highlighted area showing primary drug resistance.

(11.03%), two (1.37%), and four patients (2.75%), respectively. Resistance to two-drug combinations was observed in two patients, one with rifampicin and ofloxacin, and the other with rifampicin and dapsone (Fig. 1).

Resistance mutations were identified in 25 of 145 cases: only ofloxacin in 16 cases (*Gly89Phe, Ala91Val, Ala91Pro, Ala89Gly, Leu91Ala, Arg91Ala, Pro91Ala*, and *Phe91Ala*), ofloxacin (*Ala89Gly* and *Thr89Gly*) in combination with rifampicin (*Asn458Leu* and *Leu436Gln*) [12] in two cases, only rifampicin in four cases (*Leu436Gln, Ala438Gln, Gly441Asp, Pro451His, Val456His, Val458Leu, Leu436Gln*, and *Leu458Pro*), dapsone (*Ala53Thr*) in combination with rifampicin (*Leu436Gln*) in one case, and only dapsone in two cases (*Ala53Thr*) (Table 1) [12].

These patients with ofloxacin-resistant leprosy are residents of adjoining blocks of the Hura and Puncha areas of Purulia; other primary-drug–resistant cases are from the Jhalda-I, Arsha, Baghmundi, Manbazar-I, and Barabazar blocks, which are very close to each other (Fig. 2).

After 35 years of MDT, the prevalence of leprosy in India has decreased remarkably; however, drug resistance is starting to emerge. After completion of the scheduled course of treatment using effective chemotherapy, there should not be any relapse of the disease. However, an occurrence of relapse indicates either incomplete treatment or the emergence of a drug-resistant strain. Hence, rate of relapse is a key factor in determining the effectiveness of a chemotherapeutic regimen. As per official reports from 130 countries and territories, the registered global prevalence of leprosy has decreased from 228,474 new cases reported in 2010 to 202,185 in 2019 [13]. However, the total number of relapse cases has increased from 1734 to 3893; India contributed 505 relapse cases in 2019 and ranks in this aspect as third, after Indonesia [12]. Various studies proved the presence of secondary and primary resistance to rifampicin and dapsone in *M. leprae* [8,11,14].

In this study, we observed that resistant strains are actively circulating in some endemic pockets of the Purulia district. Our study reports the presence of ofloxacin-resistant *M. leprae* in new MB cases of leprosy. We found the highest number of resistant cases for ofloxacin (10%), followed by rifampicin (4%) and dapsone (2%).

Ofloxacin, a fluorinated carboxyquinolone belongs to the fluoroquinolones group of antibiotics and is a derivative of the first quinolone, nalidixic acid [15]. Quinolone resistance-determining region (QRDR) has been targeted for the primer designing of ofloxacin [4]. The target of fluoroquinolones is DNA gyrase, preventing replication of bacteria [16,17]. The findings from this study revealed the presence of *M. leprae* strains resistant to ofloxacin in new cases of leprosy. One reason for finding a high number of ofloxacin-resistant cases may be explained by the fact that fluoroquinolones are widely used for the treatment of infectious diseases in adults, such as chlamydia, skin and soft tissue infections, streptococcus, and other common bacterial infections [18]. Patients with medical histories of ofloxacin administration for diseases other than leprosy may develop resistance over time. We already reported cases of secondary drug resistance to rifampicin, ofloxacin, and dapsone from the same areas of the Purulia district [9,11]. Therefore, it is justifiable to mention that ofloxacin-resistant *M. leprae* released from these cases might have infected a naïve contact population and emerged as a case of primary resistance.

It is likely that resistant strains are actively circulating in some endemic regions of Purulia. Drug resistance may be due to the transmission of a mutant strain of *M. leprae* to patients, causing primary resistance, or to a mutation in a wild strain of *M. leprae* during treatment, causing secondary resistance. The risk of their transmission to a healthy population cannot be ignored. Single Nucleotide Polymorphism (SNP) subtyping was already done from the same area from 2013 onwards and we have not found any association between SNP subtypes and drug susceptibility [10,19]. New leprosy patients with primary resistance to ofloxacin, a second-line drug for the treatment of leprosy [4], may impede the progress of the National Leprosy Elimination Programme (NLEP) program. For the program to succeed, there is an urgent need to periodically monitor trends in drug resistance by setting up a centre for active vigilance that finds resistant cases and enables the immediate administration of alternate therapies. Increasing resistance rates point out the need to formulate new strategies that will succeed in arresting the spread of resistant *M. leprae* strains.

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Ethical approval

Ethical clearance for this study was approved by the The Leprosy Mission Trust India Ethical Committee, held on August 29, 2014, under the regulations of the Indian Council of Medical Research. Written informed consent was obtained from all subjects before collection of biological samples.

Declaration of Competing Interest

None declared.

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